ESHA HOODA

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Seeking Software Engineering, Machine Learning & Data Science Internships for Summer 2027

EDUCATION

Bachelor of Science, Data Science, Minors in Statistics & Computer Science, Public Health Domain Elective - San Jose State University Dec 2027

Coursework: **Machine Learning**, **Artificial Intelligence**, Statistics & Probability, Data Visualization, Data Structures & Algorithms, Database Management, Advanced **Python**, OOP, Linear Algebra, Discrete Math, UNIX, Technical Writing, R | Planned: Computer Vision, Social Network Analysis, Stats for Bioinformatics, Contemporary Environmental Health Issues, Health Issues in a Multicultural Society

Certificate: Fundamental AI Algorithms and Applications in Computer Science (expected completion: May 2027)

SKILLS & CERTIFICATIONS

Technical Skills: Python, SQL, PySpark, ETL, **PyTorch**, **TensorFlow/Keras**, **HuggingFace Transformers**, **Scikit-learn**, GCP, Azure, Git, Flask, MySQL, Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI, Docker, JavaScript, UNIX, Java, **C++**

AI Tools & Applications: Anthropic Claude API (tool-calling, **agent** design), **Claude Code**, **GitHub Copilot**, ChatGPT, Gemini, Perplexity, Grok AI (research, task automation, prompt engineering)

Certifications: Python, Machine Learning, Artificial Intelligence (IIT Bombay & IIT Madras), Code in Place (Stanford), IRB Research Training, CRLA Level 1 & 2

Languages: English (Native), Hindi (Native)

RESEARCH EXPERIENCE

Diffusion-Based Fluorescence Microscopy Super-Resolution (SR3), with Dr. Tahir Bachar Issa Sep 2026

- Started a research collaboration applying **diffusion models** (**SR3**, Image Super-Resolution via Iterative Refinement) — a **deep neural network** approach — to video-rate fluorescence microscopy super-resolution, replacing the GAN-based approach from a 2020 IEEE BIBE paper as the project baseline

AI Upskilling Research Program, Stanford SALT Lab Sep 2026

- Selected for a Stanford SALT Lab-affiliated **AI** upskilling research program; building a tool-using AI agent and a RAG-based document-search system, then hardening both against adversarial attack as part of an intensive hands-on curriculum
- Developing an evaluation suite and a **CI** regression gate to catch **AI** system regressions before they ship, running three-**model** comparisons, and deploying a **model** server into a secured private **cloud** network with no public access

Student Researcher (Advisor: Dr. Cassandra Paul), Research Foundation, SJSU May 2025 - Present

- Applied Principal Components Analysis (PCA) and Factor Analysis (**scikit-learn**) to an **11-item** Likert survey to validate three subscales (Identity & Belonging, Academic Efficacy, Navigation & Resources)
- Conducting independent research quantifying the LA model's impact on student outcomes (focus: gender equity and underrepresented groups); **manuscript underway** for conference proceeding submission, with findings communicated via data visualizations and written reports
- Building and maintaining research **databases**; performed **feature engineering**, **hyperparameter tuning**, and cleaning on ~**60,000** rows of messy institutional data for statistical analysis using **Python** and hierarchical linear modeling (HLM)

PROJECT EXPERIENCE

Trust & Safety Risk Pipeline Jul 2026

Tools Used: Python, SQL, PySpark, Airflow, dbt, BigQuery, GCP, **XGBoost**, **HuggingFace Transformers**, **PyTorch**, OpenCV, Docker, FastAPI

- Built a multimodal **AI** risk-detection pipeline combining tabular fraud signals, **NLP**-based content classification (**HuggingFace Transformers**), and CV-based image risk detection into a unified **model** score, A/B-tested against a baseline **model** for accuracy

AI Job Application Tracker Agent Aug 2026

Tools Used: Python, Flask, SQLite, **Anthropic Claude API (tool-calling)**, JavaScript, HTML/CSS, RAG (embeddings + vector search)

- Built a full-stack **agent** that acts on the user's behalf rather than just answering questions: given a job posting, it autonomously decides which tools to call (parse the posting, fetch the resume, score fit, log the result) using Anthropic's Claude tool-calling API directly, not a fixed pipeline
- Implemented a five-tool **agent** architecture with a persisted conversation log for memory across turns, backed by a **Flask** API and SQLite **database**; wrote the scoring prompt to avoid inflating fit scores, and built a **RAG** layer (embeddings + vector search)

Gaming Stats Dashboard & Pricing Models Dec 2025

Tools Used: Python, NumPy, Pandas, **Scikit-learn**, Seaborn, Matplotlib, Git

- Developed a **classification model** pipeline (Logistic Regression, Random Forest, SVM) on **20K+** games across Steam, PlayStation, and Xbox; diagnosed a **2:1** class imbalance masking true model performance via per-class recall analysis, revealing models caught only **16-18%** of high-price games despite **69%** overall accuracy

Stock Volatility Prediction Pipeline May 2026

Tools Used: Python, PySpark, SQL, ETL, Pandas, Matplotlib, Seaborn, GCP, UNIX, Git

- Engineered a distributed data pipeline (PySpark on GCP) processing **600,000+** daily trades across **500+** stocks (**2019-2023**); RandomForest feature importance identified prior-day intraday range as the dominant volatility predictor (importance: **0.837**)

LegalBee, Responsible Computing Club, SJSU Sep 2026

Tools Used: Python, RAG (embeddings + vector search, in development), LLM prompting, NLP classification (planned)

- Leading development of **LegalBee**, a jurisdiction-aware RAG copilot answering legal questions from government, court, and legal-aid sources with citations and explicit confidence signaling, rather than a generic legal chatbot
- Scoping an initial California tenant-law vertical with a small team; architecting for a source-transparency panel and deadline extraction, with a planned **ML** component (legal-topic classification, jurisdiction detection, retrieval reranking) and retrieval-accuracy/hallucination-rate evaluation metrics designed in from the start

Recipe Management Database System Dec 2025

Tools Used: MySQL, Flask, JavaScript, Flutter, Git

- Architected and implemented a full-stack CRUD application on a relational MySQL **database** with normalized schemas, indexed queries, and RESTful API integration; implemented synthetic data generation (~**30K** rows) and built analytics for ingredient frequency and cost summaries

TECHNICAL EXPERIENCE

Deep Learning Intern, National University of Singapore & Hewlett Packard Enterprise Dec 2022 - Jan 2023

- Designed and implemented **ML** and **deep neural network** models including **ANN, RNN, GAN, SVM**, **NLP**, and time series architectures; gained hands-on experience with **Azure** deployment pipelines
- Trained and optimized a posture detection **model** achieving **93%** accuracy through **neural network architecture** tuning and iterative experimentation

Learning Assistant, Peer Connections, SJSU Jan 2025 - Jan 2026

- Mentored **250+** students in math and CS through strong communication and cross-functional collaboration with professors and peer tutors to co-design equity-focused in-class learning activities; earned CRLA Level 1 & 2 certification

LEADERSHIP EXPERIENCE

Outreach Lead, Peer Connections, SJSU May 2025 - Jan 2026

- Scaled a major initiative from **0** to **3,000** participants in **2** months by redesigning internal processes and building **SQL-based** tracking systems while coordinating a team of **120+**

Vice President, Math Club, SJSU Oct 2025 - Mar 2026

- Organized weekly problem-solving events and partnered with faculty to promote engagement in mathematics

AWARDS & ACHIEVEMENTS

- State Rank **2** (**3** consecutive years) in IEO, with international ranks of **232**, **474**, and **660** among millions of students across **70+** countries
- Top **150** nationwide, National Children's Science Congress (Govt. of India; selected from **1 million** students)